

IMO Training – Number Theory

Level 1 Assignment 1

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Instructions:

- Write your solutions legibly on separate sheets of paper.
- Each question carries 7 marks.
- Complete solutions with justified arguments are required for full credit in every question.
- Both paper and electronic copies are accepted.
- Deadline for submission: 10 August, 2019

1. Show that for every positive integer n , $121^n - 25^n + 1900^n - (-4)^n$ is divisible by 2000.
2. It is given that $34! = 295232799cd96041408476186096435ab000000$. Determine a, b, c, d .
3. Find all two-digits integers N for which the sum of the digits of $10^N - N$ is divisible by 170.
4. Let $a_1 = 20, a_2 = 19$. For $n \geq 1$, define a_{n+2} to be the remainder of $a_n + a_{n+1}$ when it is divided by 100. What is the remainder when $a_1^2 + a_2^2 + \cdots + a_{2019}^2$ is divided by 8?
5. Find all positive integers whose squares end in three 4's. Are there any perfect squares that end with four 4's?
6. Find a six-digit integer n such that n is a perfect square, and the number formed by the last three digits of n is exactly one greater than the number formed by the first three digits of n .
7. Find four distinct positive integers whose product is divisible by the sum of every pair of them. Can you find a set of five or more numbers with the same property?
8. Find all positive integers a, b, c such that $a!b! = a! + b! + c!$.
9. Let p, q, r be prime numbers. It is given that $p|qr - 1, q|pr - 1, r|pq - 1$. Find all possible values of pqr .

Classwork

30 July:

1.1 Divisibility Q 4,8,12

2.1 Congruence Q 29,32

1 August:

2.1 Congruence Q 14,15,38,39,40,44